

Recommendation Report: Mine Site Selection for Northern BC

Prepared For: Copper-nicus Mining Ltd.

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1. Scope Of Work

Mining sites in BC, particularly in the northern region, face various logistical and environmental constraints due to their remote locations and sensitive ecosystems. Ensuring safety, efficiency and economic viability is essential for a successful mining project in the Nicus mountain range. Our team thoroughly examined 3 copper deposits independently to determine which site is feasible.

Our team narrowed in on specific evaluation criteria: community benefits, impacts of mines on the area, preservation of cultural sites, profit and efficiency of the mine, how this mine will increase infrastructure and help the community expand, and how resiliency and versatility. These criteria enabled us to dive into the technical, environmental, social and economic implications of each possible option.

Several constraints may have influenced the outcomes of our consultation and analysis. These include time limitations, limited stakeholder interview footage, and incomplete information regarding deposit size, quality, and accessibility, as well as limitations related to mine type, location, and size options. As a result, our findings are inherently shaped by these factors and should be considered when planning a final project. It is also important to note that this report does not take into account mine operations, public relations beyond the local population, or external market forces such as international price fluctuations in copper concentrate.

Furthermore, our analysis assumes the following:

- All stakeholder consultations were performed honestly and in good faith, with a sufficiently representative sample of local residents.
- No attempts were made by local residents to prevent or intentionally interfere with the mine's construction.
- Explorations of the mine locations were conducted appropriately and accurately.

The purpose of our report was to analyze the 3 different mine sites, and determine the most effective solution. We have come to the conclusion that Deposit 2 outperforms the other deposits in our specific evaluation criteria and is the best option for the next Copper-nicus mine site.

2. Engagement, Learning, & Synthesis Strategies

Continuous engagement with stakeholders and right holders is necessary to ensure sustainability concerns are addressed. An engagement strategy will be put in place to ensure stakeholder feedback is incorporated into project planning, and that it aligns the project with the priorities of local communities and Indigenous rights holders.

Local Indigenous communities will play an influential role in the development of this mine. In order to engage with the local indigenous communities we will implement formal consultation meetings throughout all phases of the mine's life cycle. Their generous participation will guide us to ensure that archaeological sites will be undisturbed and traditional land practices are protected.

Community engagement will also include regular public information sessions in the form of open houses. These meetings will be to update the community on the overall progress, allow members to voice their concerns and enable them to provide feedback on environment managing strategies and infrastructure plans. Maintaining communication with the community will help gain the trust of the people and implement the locals priorities into the project.

The success of this mine will be measured in terms of technical achievement and community acceptance. Measurement of technical success will include safe operations of the mine, satisfaction of policy makers and compliance with environmental regulations, and meeting profit and production targets. Community acceptance will be measured through stakeholder satisfaction; by engaging with stakeholders, their positive feedback will tell us we are flourishing.

Some challenges we may face are unforeseen community concerns regarding environmental impacts, interferences in cultural land, or misalignment between stakeholder values and priorities. In order to prepare for these challenges, they will be addressed through consultation early on, and constant environmental assessment and regular consultation with those in the local community will help us tackle the concerns regarding the environment. By implementing feedback into the planning phase, and maintaining open communication and receiving that constant feedback, the team will aim to build long term community support during the development of the selected mining deposit.

3. Analysis of Mine Sites

3.1 Deposit 1

3.1.1 Description of the Deposit

- **Key Features:** Deposit 1 is a deep, narrow deposit containing high-grade copper ore. The rock quality is variable, and the site is highly remote, currently requiring helicopter access.
- **Mining Method:** Due to its depth, specialised underground mining methods are required.
- **Stakeholder and Rights Holder Impacts:** The extreme remoteness minimises direct disturbances (such as noise or dust) to the local town and First Nations. However, the deep underground environment introduces significant health and safety risks for the workers.
- **Resiliency:** The resiliency of this site is poor. Heavy rainfall would require expensive drainage and water-pump systems, adding to both operational costs and GHG emissions. Furthermore, due to the high initial infrastructure costs, if the ore grade turns out lower than expected, it would severely impact the mine's overall profitability. Lastly, losing specialised underground workers would disproportionately affect operations compared to an open-pit mine.
- **CLD Relationships:** As shown in the Causal Loop Diagram (CLD) (see Appendix), the remote location requires substantial initial "Mine infrastructure" investment, which negatively impacts short-term "Mine profitability." However, the high-grade ore results in less "Waste production," which creates a positive relationship with the "Health of local ecosystem and preservation of landform".

3.1.2 Environmental Impact of the Solution

- **Environmentally sustainable?** It is relatively sustainable in terms of land use. Because it is an underground mine with high-grade ore, it will generate less waste rock and require less water use during processing compared to open-pit alternatives.
- **Requirements to meet environmental criteria outlined by stakeholders and rights holders:** Although there are no major water bodies nearby, the annual elk migration passes through the

area. To meet the stakeholders' strict criteria for the preservation of the land, robust wildlife mitigation measures must be designed and implemented to protect the elk route.

3.1.3 Economic Viability and Sustainability

- **Economic impact on stakeholders and rights holders:** The estimated break-even period is lengthy at 9-12 years due to the 3 years needed to build an access road and 2 years for infrastructure (camp, power). However, our preliminary block model optimization indicates a potential profit of approximately \$6.95 billion. While it meets the overarching goal of being a "Profitable mine", the remote nature means there will be fewer direct infrastructure benefits (like shared local roads) for the nearby town.

3.1.4 Social Sustainability

- **Potential social impact on the local town and First Nations groups:** The physical distance ensures that the daily lives and ecotourism businesses of the First Nations and the town are largely undisturbed.
- **Social sustainability of our option:** The primary social sustainability concern is occupational health and safety, as deep underground mining limits access to emergency health services.
- **How does it meet the social criteria outlined by the stakeholders and rights holders?** It meets the "respectful operations" criteria by naturally eliminating community disturbances. However, it scores lower on "community benefits" because the specialised nature of the work and the remote location may limit local employment opportunities.

3.1.5 Other Considerations

- Based on our Weighted Decision Matrix (WDM), Deposit 1 received the lowest overall score (4.95). The major factors dragging down its score were its poor resiliency (scoring 2/10) and the lack of community benefits (scoring 1/10).

3.2 Deposit 2

3.2.1 Description of the Deposit

- **Key Features:** Deposit 2 is moderately remote, located near Nicus River and Nicus Lake, opposite to the local community. Ore grade increases, while rock stability decreases with depth.
- **Mining Method:** An underground mine provides access to better grade ore as depth increases, while minimizing the waste rock removed (34,830,000 tonnes removed).
- **Stakeholder and Rights Holder Impacts:** Improvements to local infrastructure will develop this area for the future. It will also provide job opportunities for Indigenous and local communities. Indigenous communities and locals have environmental concerns related to the proximity to the river and lake, therefore mitigation strategies will need to be implemented and agreed upon. Policy makers will require frequent development updates to ensure the mine is up to code. The construction company will provide more specialized equipment as mining progresses.
- **Resiliency:** The resilience of Deposit 2 is moderate. Because the mine is moderately remote, only some support is provided by local communities in the case of a disturbance. The proximity to bodies of water increases the risk of environmental damage and worsens resilience.
- **CLD Relationships:** As depicted in Appendix A, an increase in "Material removal", caused by the improved grade as depth increases, leads to an overall increase in "Mine profitability". The increase in "Processed material" will lead to a decrease in the "Health of local ecosystem and preservation of landform", depicting the risks related to the nearby river and lake.

3.2.2 Environmental Impact of the Solution

- **Environmentally sustainable?** Because it is an underground mine, the land use is quite sustainable. However, due to Deposit 2's proximity to the nearby Nicus River and Lake, significant environmental precautions must be taken to preserve these bodies of water.
- **Requirements to meet environmental criteria outlined by stakeholders and rights holders:** For tailings, a nearby valley can be utilized as a storage area. Control methods such as installing a tailings dam liner, dewatering of tailings to minimize the potential for runoff water, and frequent water testing should be implemented. Frequent consultation with local Indigenous communities is necessary to ensure their satisfaction of the treatment of the land.

3.2.3 Economic Viability and Sustainability

- **Economic impact on stakeholders and rights holders:** Stakeholders and rights holders addressed the importance of providing jobs to local communities, and bringing in revenue to give back to the community. Deposit 2 excels in this area, with a profit of \$7,849,912,500. Some of this profit will aid in environmental preservation costs, while also providing for the community. Job opportunities will be provided for those who are willing to commute to this moderately remote site. While mining costs per tonne will increase as extraction progresses deeper, as there is a need for more advanced equipment and safety precautions, these costs will likely be offset by the higher grade of copper at these depths. The estimated break even period is 7-14 years—the greater uncertainty coming from the uncertainty around the rock stability.

3.2.4 Social Sustainability

- **Potential social impact on the local town and First Nations groups:** In order for operation on the mine to begin, a bridge, access road, accommodation camp, and power supply will need to be installed. This will increase access to the site for local communities and mine operators.
- **Social sustainability of our option:** This construction will take approximately 1.5 years. During this time, consistent consultation and communication with local communities will continue.
- **How does it meet the social criteria outlined by the stakeholders and rights holders?** The social sustainability of Deposit 2 is highly reliant on the communication between mine operators and key stakeholders such as Indigenous and local communities. More information regarding our ongoing communication plan is included in Section 2 of this report.

3.2.5 Other Considerations

- Based on our WDM in Appendix C, the Deposit 2 underground mine earned a total score of 6.19. Strengths are its profitability and opportunity for improvements to local infrastructure, and drawbacks are risk of environmental damage, and low resilience.

3.3 Deposit 3

3.3.1 Description of the Deposit

- **Key Features:** Deposit 3 is a large, shallow deposit with an extensive lateral reach, containing low-grade copper ore. It is highly accessible, located near an existing highway and close to the local town.
- **Mining Method:** The deposit's shallow nature makes it cost-effective for open-pit mining.
- **Stakeholder and Rights Holder Impacts:** Being very close to civilization improves health and safety access and offers substantial potential jobs for the community. However, the town will be highly affected by mining operations, requiring careful consultation.

- **Resiliency:** Open-pit resiliency presents unique challenges. Heavy rainfall could lead to a loss of pit wall stability, requiring shallower angles that generate even more waste rock, or cause flooding at the base of the pit. Additionally, if an unknown cultural site is discovered during excavation, it is exceedingly difficult to mine around it while preserving the land.
- **CLD Relationships:** The CLD illustrates that proximity to the town strongly drives "Local employment" and "Local economic health." However, the massive "Material removal" required for a low-grade open pit drastically increases "Waste production," which can negatively impact the "Social license to operate" if dust and landscape changes upset the community.

3.3.2 Environmental Impact of the Solution

- **Environmentally sustainable?** The environmental footprint is significant. The low-grade ore means massive amounts of waste rock will be produced, and high water use is required for processing.
- **Requirements to meet environmental criteria outlined by stakeholders and rights holders:** Dust control, landform impacts, and land use will be top priorities. Furthermore, the most suitable valley for the tailings dam trends north-south toward an archaeological site. To meet the criteria for environmental and cultural preservation, the tailings storage facility must be impeccably engineered to guarantee zero seepage or risk of failure at the site.

3.3.3 Economic Viability and Sustainability

- **Economic impact on stakeholders and rights holders:** This site requires minimal infrastructure investment due to its proximity to the town, leading to a short break-even period of 5-8 years. Our block model estimates a profit of approximately \$5.68 billion. This option strongly meets the "Community benefits" criteria by providing accessible local jobs and funnelling money directly back into the local economy.

3.3.4 Social Sustainability

- **Potential social impact on the local town and First Nations groups:** The economic integration with the town is a major positive. However, an archaeological site of cultural significance to both First Nations is situated approximately 50 km north.
- **Social sustainability of our option:** There is a high risk of strong opposition regarding the use of the nearby valley for a tailings dam.
- **How does it meet the social criteria outlined by the stakeholders and rights holders?** While it scores perfectly on community benefits (8/10 on the WDM), meeting the strict criteria for the "Preservation of cultural sites" will require unprecedented collaboration, transparent monitoring, and potentially costly engineering alternatives (such as dry-stack tailings) to secure the First Nations' consent.

3.3.5 Other Considerations

- Deposit 3 placed second in our WDM with a score of 5.40. While its resilience score is excellent (9/10), it scored very poorly on environmental preservation (2/10) and cultural site preservation (4/10), making it a high-risk option from a stakeholder engagement perspective.

4. Final Selection and Justification

Determining Criteria, Weights, and Scores

- Key stakeholders and their weights were determined based on power, legitimacy, and urgency.
 - Local Communities (15%), Local Indigenous Communities (25%), Policy Makers (35%), Mining Company (20%), and Construction Company (5%).

- Based on the stakeholder and rights holder interview video, the evaluation criteria were determined. Using this video and external research, a stakeholder WDM was created to attribute a weight to each evaluation criteria. Detailed scoring can be found in the WDM in Appendix C.
 - Community Benefits (16%), Preservation of the Environment (18%), Preservation of Cultural Sites (13%), Profitability (18%), Improvement to Local Infrastructure (12%), and Resilience (23%).

WDM Outcomes

As seen in Appendix C, the following final scores, strengths, and weaknesses were identified.

- Deposit 1 Underground - Score: 4.95/10
 - Strengths: Preservation of cultural sites, improvement to local infrastructure
 - Weaknesses: Poor resilience, lack of community benefits
- Deposit 2 Underground - Score: 6.19/10
 - Strengths: Profitability, improvements to local infrastructure
 - Weaknesses: Poor environmental preservation, sub-optimal resilience
- Deposit 3 Open Pit - Score: 5.40/10
 - Strengths: Resilience, community benefits
 - Weaknesses: Poor environmental preservation, profitability

Streamlined Life Cycle Assessment (SLCA) Outcomes

The SLCAs detailed in Appendix B conclude that Deposit 1 and Deposit 2 are equally viable with a total score of 25, followed by Deposit 3 with a score of 21.

- Deposit 1 excelled Waste Generation and Water Use, with drawbacks surrounding Health and Safety.
- Deposit 2 had good scores for Emissions, Waste Generation, and Water Use. It performed slightly better in Health and Safety compared to Deposit 1.
- Deposit 3 performed best in Health and Safety, as it is an open pit mine, but performed poorly in Emissions and Waste Generation due to larger equipment needed and waste rock amounts.

Final Recommendation: Deposit 2 Underground Mine

Based on our assessment of all 3 potential mine sites, we conclude that the most viable option is an underground mine at Deposit 2 due to its high WDM performance and strong SLCA scores. The strengths of Deposit 2 are its high profitability due to higher grade ore at a lower depth, and opportunity for improvements to infrastructure. The high profitability can be used to give back to local communities, in addition to providing job opportunities. This satisfies the needs expressed by Indigenous communities and local communities. Because this area is moderately remote, improvements made to local infrastructure now will benefit communities in the future.

The environmental risk of the proximity to Nicus River and Nicus Lake must be monitored. Precautions must be taken to preserve these bodies of water such as installing a tailings dam liner, dewatering of tailings to minimize the potential for runoff water, and frequent water testing. The uncertainty related to the worsening rock stability as depth increases must also be managed. This will require more advanced equipment and more safety measures, which will increase costs. These costs will likely be offset by the higher grade ore at these depths, but should be managed.

5. Appendices

Appendix A: Causal Loop Diagram

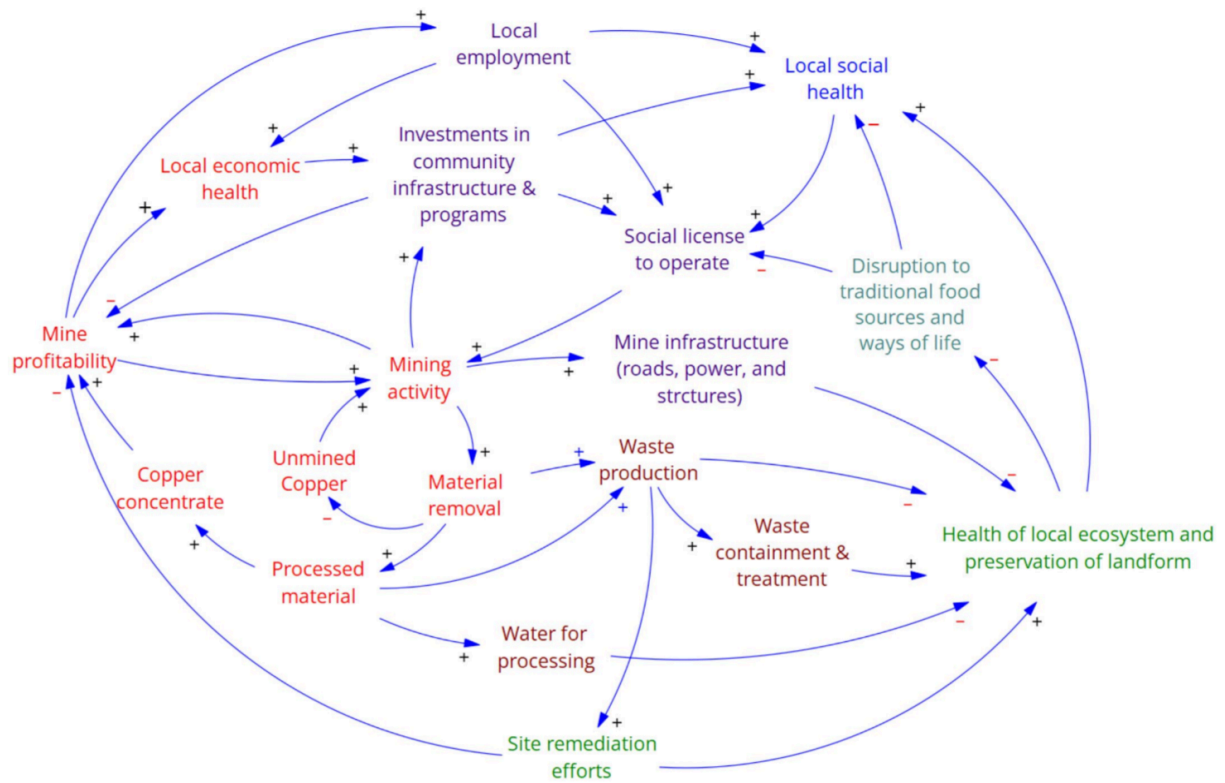


Figure 1: Causal Loop Diagram for a Mining Operation

Appendix B: Streamlined Life Cycle Assessment (SLCA) for Each Deposit

Deposit 1 - Underground

	Development	Operation	Closure	Total
Health and safety	0	1	1	2
Emissions	2	3	2	7
Waste generation	2	3	3	8
Water use	3	3	2	8
Total	7	10	8	25

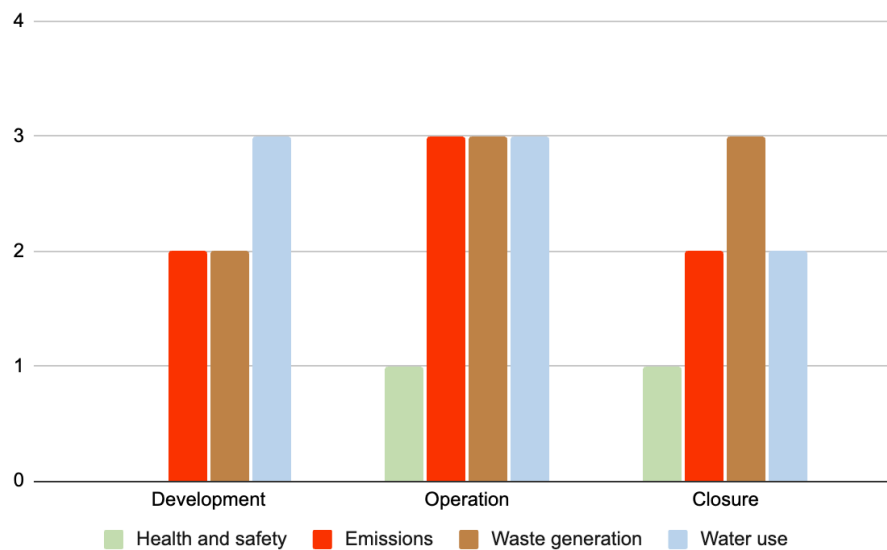


Figure 2: Graphical Form of SLCA for Deposit 1 Underground

Deposit 2 - Underground

	Development	Operation	Closure	Total
Health and safety	1	1	1	3
Emissions	2	3	2	7
Waste generation	3	2	3	8
Water use	3	2	2	7
Total	9	8	8	25

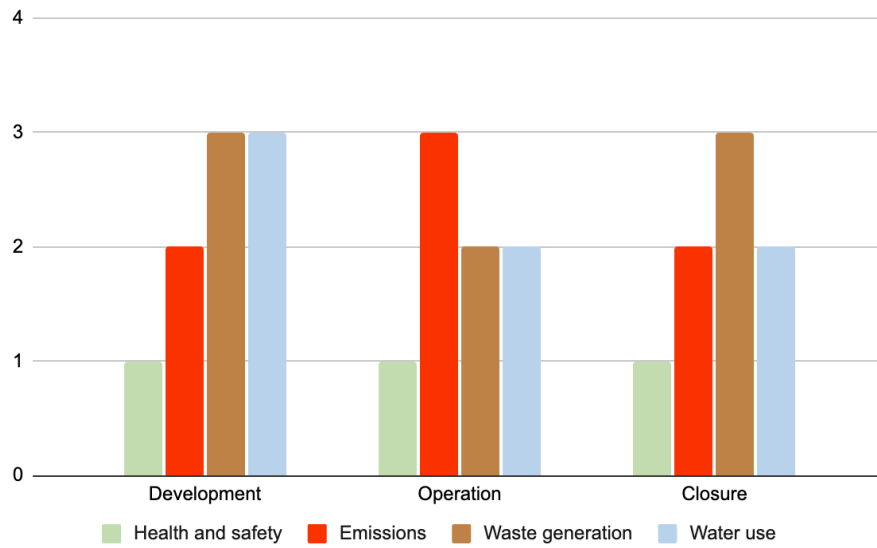


Figure 3: Graphical Form of SLCA for Deposit 2 Underground

Deposit 3 - Open Pit

	Development	Operation	Closure	Total
Health and safety	2	2	3	7
Emissions	1	1	1	3
Waste generation	2	1	1	4
Water use	3	2	2	7
Total	8	6	7	21

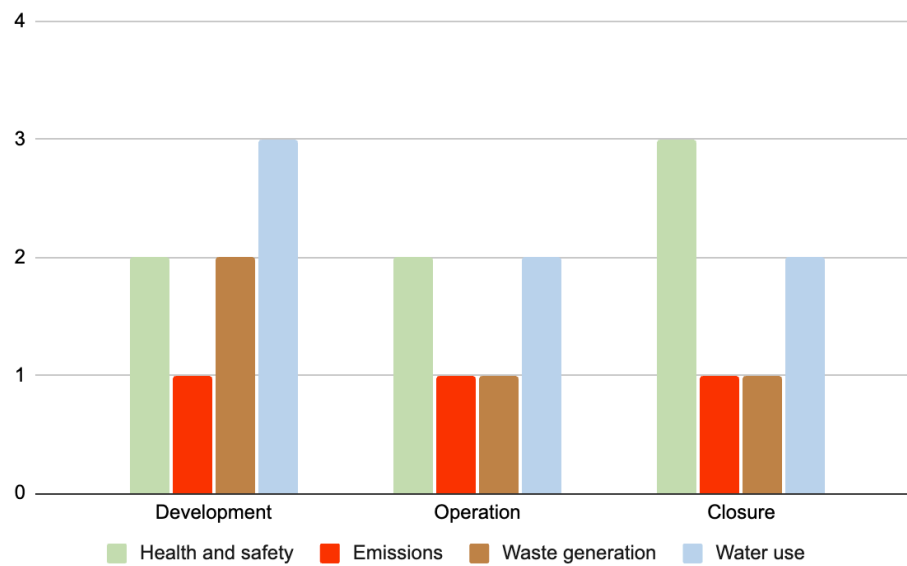


Figure 4: Graphical Form of SLCA for Deposit 3 Open Pit

Appendix C: Weighted Decision Matrix (WDM)

a. Stakeholder WDM

Stakeholders	Stakeholder Weight	Community benefits		Preservation of the environment		Preservation of cultural sites		Profitability		Improvement to local infrastructure		Resilience	
		Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted
Local Communities	15%	10	1.5	8	1.2	4	0.6	9	1.35	10	1.5	9	1.35
Local Indigenous Communities (Rights Holders)	25%	10	2.5	10	2.5	10	2.5	9	2.25	8	2	10	2.5
Policy Makers	35%	5	1.75	7	2.45	5	1.75	3	1.05	1	0.35	7	2.45
Mining Company	20%	1	0.2	3	0.6	1	0.2	10	2	3	0.6	10	2
Construction Company	5%	2	0.1	2	0.1	0	0	8	0.4	6	0.3	8	0.4
Totals:			6.05		6.85		5.05		7.05		4.75		8.7
Weightings for each criteria			15.7%		17.8%		13.1%		18.3%		12.4%		22.6%

b. Main WDM

Evaluation Criteria	Weight	Deposit 1 (underground)		Deposit 2 (underground)		Deposit 3 (open pit)	
		Score	Weighted	Score	Weighted	Score	Weighted
Community benefits	15.7%	1	0.16	6	0.94	8	1.26
Preservation of the environment	17.8%	5	0.89	4	0.71	2	0.36
Preservation of cultural sites	13.1%	9	1.18	8	1.05	4	0.53
Profitability	18.3%	7	1.28	10	1.83	4	0.73
Improvement to local infrastructure	12.4%	8	0.99	6	0.74	4	0.49
Resilience	22.6%	2	0.45	4	0.91	9	2.04
Totals:			4.95		6.19		5.40

Appendix D: Mine Block Models

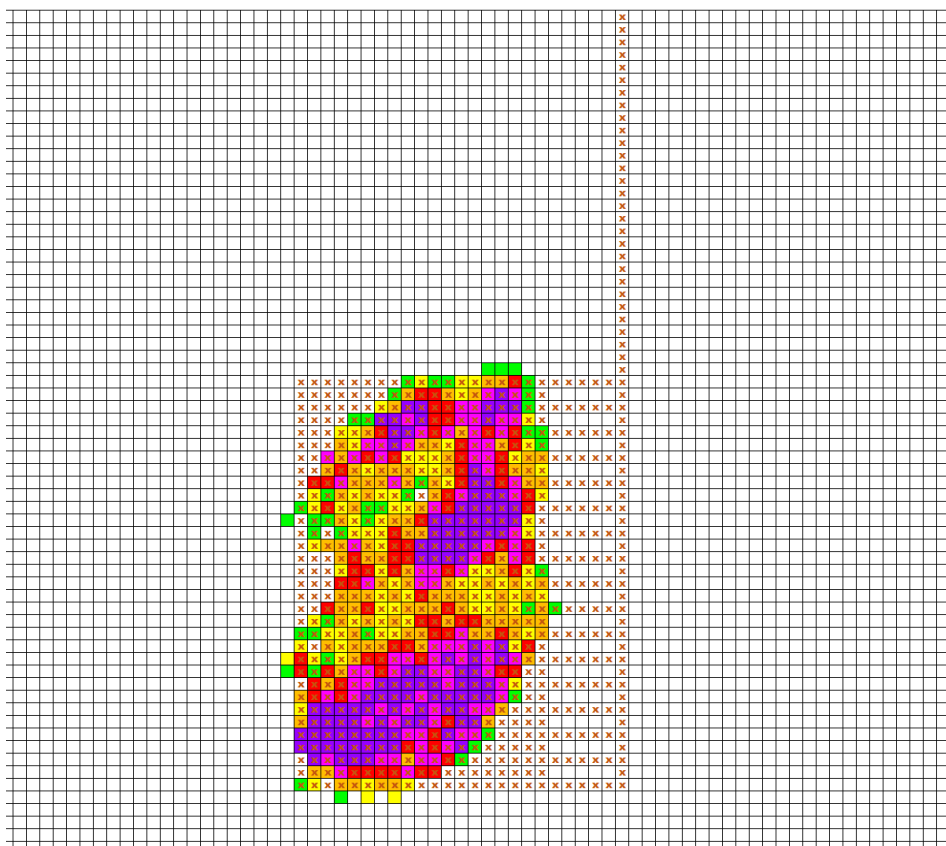


Figure 5: Deposit 1 Underground Block Model

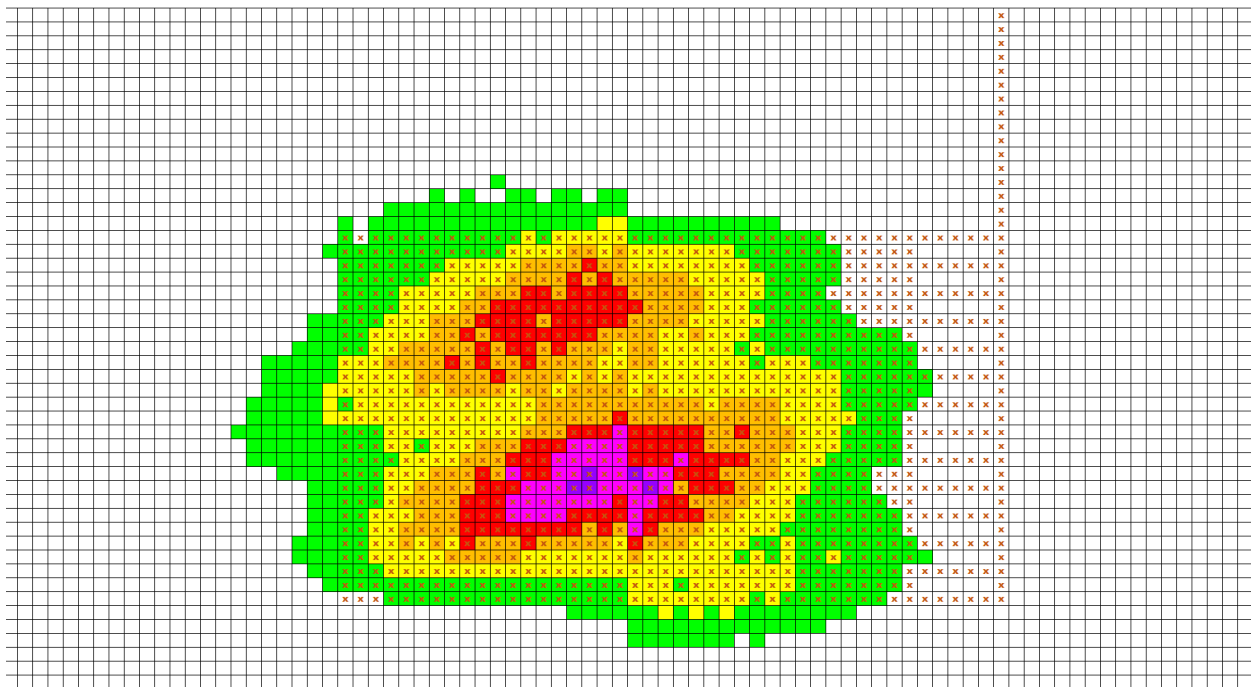


Figure 6: Deposit 2 Underground Block Model

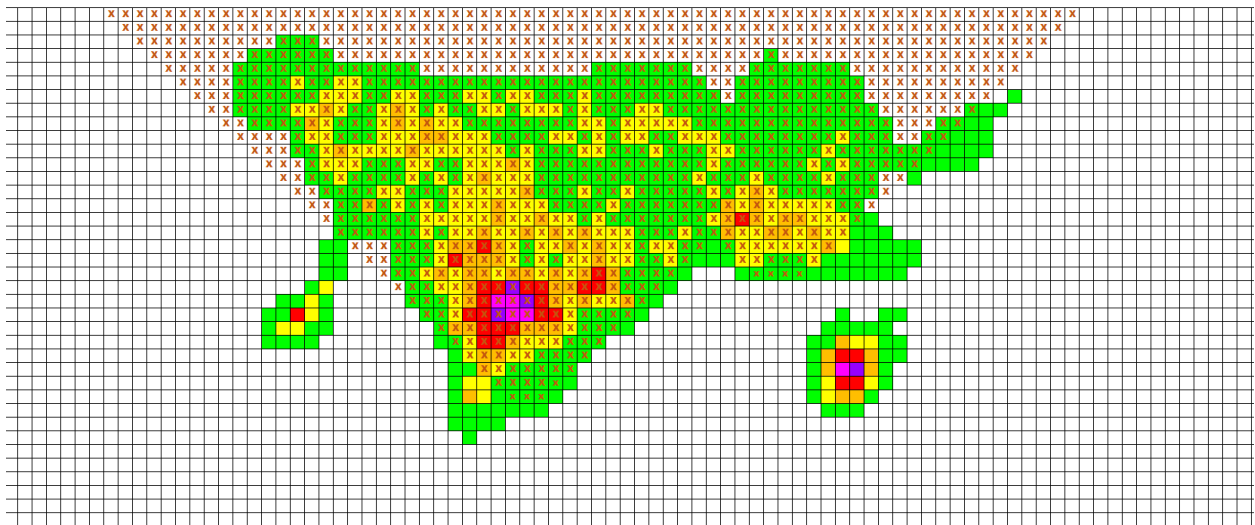


Figure 7: Deposit 3 Open Pit Block Model

6. GenAI Disclosure Statement

No generative AI was used in the creation of this report.